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INTEGRATED CIRCUIT PACKAGES AND METHODS OF FORMING THE SAME

Abstract

A method of forming an integrated circuit package includes following operations. A padding layer is formed on a portion of a carrier. A first semiconductor die is placed on the padding layer and a second semiconductor die is placed on the carrier. The first semiconductor die and the second semiconductor die are encapsulated with a first encapsulation layer. A first redistribution layer structure is formed over the first semiconductor die, the second semiconductor die and the first encapsulation layer. A third semiconductor die is placed on the first redistribution layer structure. The third semiconductor die is encapsulated with a second encapsulation layer. A second redistribution layer structure is formed over the third semiconductor die and the second encapsulation layer. The carrier is debonded. The padding layer is removed, and therefore, a recess is formed in the first encapsulation layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional application of and claims the priority benefit of a prior application Ser. No. 17/683,377, filed on Mar. 1, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

[0002] In recent years, the semiconductor industry has experienced rapid growth due to continuous improvement in integration density of various electronic components, e.g., transistors, diodes, resistors, capacitors, etc. For the most part, this improvement in integration density has come from successive reductions in minimum feature size, which allows more components to be integrated into a given area. Although the existing integrated circuit packages have been generally adequate for their intended purposes, they have not been entirely satisfactory in all respects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 to FIG. 8 are cross-sectional views of a method of forming an integrated circuit package in accordance with some embodiments.

[0004] FIG. 9 to FIG. 13 are cross-sectional views of a method of forming an integrated circuit package in accordance with some embodiments.

DETAILED DESCRIPTION

[0005] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below for the purposes of conveying the present disclosure in a simplified manner. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a second feature over or on a first feature in the description that follows may include embodiments in which the second and first features are formed in direct contact, and may also include embodiments in which additional features may be formed between the second and first features, such that the second and first features may not be in direct contact. In addition, the same reference numerals and/or letters may be used to refer to the same or similar parts in the various examples the present disclosure. The repeated use of the reference numerals is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0006] Further, spatially relative terms, such as “beneath”, “below”, “lower”, “on”, “over”,

“overlying”, “above”, “upper” and the like, may be used herein to facilitate the description of one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0007] The thermal control may protect the devices or chips in the integrated circuit package from a thermal damage. Thicker silicon is desired because it can provide larger thermal mass for a high performance chip. However, the total Z height of the package constrains the silicon thickness. In the disclosure, with novel arrangement of multiple chips in different levels or tiers, the heat dissipation efficiency is effectively improved by increasing silicon thickness of the high-performance die without increasing the total height of the multi-chip semiconductor package.

[0008] FIG. 1 to FIG. 8 are cross-sectional views of a method of forming an integrated circuit package in accordance with some embodiments. It is understood that the disclosure is not limited by the method described below. Additional operations can be provided before, during, and/or after the method and some of the operations described below can be replaced or eliminated, for additional embodiments of the methods.

[0009] Although FIG. 1 to FIG. 8 are described in relation to a method, it is appreciated that the structures disclosed in FIG. 1 to FIG. 8 are not limited to such a method, but instead may stand alone as structures independent of the method.

[0010] Referring to FIG. 1, a carrier C1 is provided with a padding layer PL formed thereon. In some embodiments, the carrier C1 is a non-semiconductor material, such as a glass carrier, a ceramic carrier, or the like. In some embodiments, the padding layer PL includes a photo-sensitive material such as polybenzoxazole (PBO), polyimide (PI), benzocyclobutene (BCB), the like, or a combination thereof, and is formed by a photolithography process. In some embodiments, the padding layer PL include an inorganic material, such as silicon oxide, silicon nitride, silicon oxynitride, or any suitable dielectric material or an insulating material, and is formed by a deposition process followed by a laser drilling process. The padding layer PL is referred to as a “polymer layer”, “dielectric layer”, a “booster layer” or an “insulating layer” in some examples. In some embodiments, the thickness or height H0 of the padding layer PL ranges from about 200 um to 500 um.

[0011] Referring to FIG. 2, through integrated fan-out vias TIV1 are formed on the padding layer PL. In some embodiments, the through integrated fan-out vias TIV1 include copper, nickel, titanium, the like, or a combination thereof, and are formed by photolithography, plating, and photoresist stripping processes. The through integrated fan-out vias TIV1 are in contact with the padding layer PL.

[0012] Thereafter, a semiconductor die 100 is placed on the padding layer P1 and a semiconductor die 200 is placed on the carrier C1. In some embodiments, the semiconductor die 100 is surrounded by the through integrated fan-out vias TIV1. In some embodiments, the semiconductor die 100 is adhered to the padding layer PL with a first die attach film (DAF) between the padding layer PL and the back side of the semiconductor die 100, and the semiconductor die 200 is adhered to the carrier C1 with a second die attach film (DAF) between the carrier C1 and the back side of the semiconductor die 200. In some embodiments, the semiconductor die 200 is separated from the padding layer PL by a distance. However, the disclosure is not limited thereto. In some embodiments, the semiconductor die 100 may be in contact with the sidewall of the padding layer PL.

[0013] Each of the semiconductor dies 100 and 200 may include a logic die, a memory die, a CPU, a GPU, an xPU, a MEMS die, a SoC die, or the like. The semiconductor dies 100 and 200 may have the same or different functions. In some embodiments, each of the semiconductor dies 100 and 200 is a SoC die.

[0014] In some embodiments, the semiconductor die **100** includes a substrate **100a**, connectors **100b** and a passivation layer **100c**. The substrate **100a** is a silicon substrate. The substrate **100a** has a device layer (not shown) formed thereon. The device layer includes a transistor, an interconnect structure, etc., and the connectors **100b** are formed over the substrate **100a** and electrically connected to the device layer. The connectors **100b** are metal pillars (e.g., copper pillars). The metal pillars include Cu, W, Ni, Sn, Ti, Au, an alloy or a combination thereof, and are formed by an electroplating process. The passivation layer **100c** is formed around the connectors **100b**. The passivation layer **100c** includes a polymer material such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), the like, or a combination thereof. In some embodiments, the surfaces of the connectors **100b** are flushed with the surface of the passivation layer **100c**.

[0015] In some embodiments, the semiconductor die **200** includes a substrate **200a**, connectors **200b** and a passivation layer **200c**. The materials and arrangements of elements of the semiconductor die **200** are similar to those of the semiconductor die **100**, so the details are not iterated herein.

[0016] The difference between the semiconductor die **100** and the semiconductor die **200** lies in the critical dimensions (e.g., the minimum line width, the minimum hole size, etc.) and chip heights (e.g., the height from the substrate to the connectors or bumps).

[0017] The critical dimension of the semiconductor die **100** is much greater than the critical dimension of the semiconductor die **200**. In some embodiments, the ratio of the critical dimension of the semiconductor die **100** to the critical dimension of the semiconductor die **200** is greater than 1.2, 1.5, 2.0, 2.5 or more. For example, the critical dimension of the semiconductor die **200** ranges from about 5 nm to 10 nm, and the critical dimension of the semiconductor die **200** ranges from about 1 nm to 4 nm. The die with the smallest critical dimension is referred to as a “high-performance die”, a “key die” or a “core die” in some examples. In the disclosure, the high-performance die is the semiconductor die **200** through the specification.

[0018] The height H2 of the semiconductor die **200** is much greater than the height H1 of the semiconductor die **100**. In some embodiments, the ratio of the height H2 of the semiconductor die **200** to the height H1 of the semiconductor die **100** is greater than 1.2, 1.5, 2.0, 2.5 or more. For example, the height H2 of the semiconductor die **200** ranges from about 200 μm to 500 μm , and the height H2 of the semiconductor die **200** ranges from about 600 μm to 1000 μm . In some embodiments, the height H2 of the semiconductor die **200** is substantially equal to the total height of the height H0 of the padding layer PL and the height H1 of the semiconductor die **100**.

[0019] In some embodiments, the ratio of the silicon substrate **200a** of the semiconductor die **200** to the silicon substrate **100a** of the semiconductor die **100** is greater than 1.2, 1.5, 2.0, 2.5 or more. Thicker silicon substrate provides larger thermal mass for the high-performance die (e.g., semiconductor die **200**), and therefore, the heat dissipation of the package is effectively reduced.

[0020] Referring to FIG. 3, an encapsulation layer E1 is formed around the semiconductor die **100**, the semiconductor die **200**, and the through integrated fan-out vias TIV1. In some embodiments, the encapsulation layer E1 is in contact with the padding layer PL and fills in the gap between the padding layer PL and the semiconductor die **200**. The encapsulation layer E1 may be formed by a molding process followed by a grinding process. In some embodiments, the surfaces of the connectors **100b** of the semiconductor die **100**, the surfaces of the connectors **200b** of the semiconductor die **200**, the surfaces of the through integrated fan-out vias TIV1 and the surface of the encapsulation layer E1 are flushed with each other.

[0021] Thereafter, a redistribution layer structure RDL1 is formed over and electrically connected to the semiconductor die **100**, the semiconductor die **200**, and the through integrated fan-out vias TIV1. In some embodiments, the redistribution layer structure RDL1 includes redistribution layers **104** embedded by polymer layers **102**. In some embodiments, each of the redistribution layers **104** includes copper, nickel, titanium, the like, or a combination thereof, and is formed by photolithography, plating, and photoresist stripping processes. In some embodiments, each of the

polymer layers **102** includes a polymer material such as polybenzoxazole (PBO), polyimide (PI), benzocyclobutene (BCB), a combination thereof or the like, and is formed by a suitable fabrication technique such as spin-coating, lamination, deposition or the like.

[0022] Still referring to FIG. 3, through integrated fan-out vias TIV2 are formed over and electrically connected to the redistribution layer structure RDL1. In some embodiments, the through integrated fan-out vias TIV2 include copper, nickel, titanium, the like, or a combination thereof, and are formed by photolithography, plating, and photoresist stripping processes.

[0023] Thereafter, a semiconductor die **300** is provided and bonded to (e.g., flip-chipped to) the redistribution layer structure RDL1. In some embodiments, the semiconductor die **300** is surrounded by the through integrated fan-out vias TIV2. The semiconductor die **300** may include a logic die, a memory die, a CPU, a GPU, an xPU, a MEMS die, a SoC die, or the like. The semiconductor dies **100**, **200** and **300** may have the same or different functions. In some embodiments, each of the semiconductor dies **100**, **200** and **300** is a SoC die.

[0024] In some embodiments, the semiconductor die **300** includes a substrate **300a**, through substrate vias **300b**, an interconnect structure **300c** and bumps **300d**. The substrate **300a** is a silicon substrate. The substrate **300a** has a transistor (not shown) formed thereon, and the interconnect structure **300c** is formed over the substrate **300a** and electrically connected to the transistor. In some embodiments, through substrate vias **300b** (also called “through silicon vias” in some examples) penetrate through the substrate **300a** and are landed on some conductive features of the interconnect structure **300c**. The interconnect structure **300c** includes conductive features embedded by dielectric layers. The conductive features include metal lines, metal vias, metal pads and/or metal connectors. In some embodiments, each conductive feature includes Cu, Al, Ti, Ta, W, Ru, Co, Ni, the like, or a combination thereof. In some embodiments, each dielectric layer includes silicon oxide, silicon nitride, silicon oxynitride, SiOC, the like, or a combination thereof. An etching stop layer may be interposed between two adjacent dielectric layers. The dielectric layers of the interconnect structure **300c** may be replaced by polymer layers or insulating layers as needed.

[0025] The bumps **300d** are formed over and electrically connected to the interconnect structure **300c**. The bumps **300d** include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps **300d** may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing. In some embodiments, the semiconductor die **300** is turned over and then bonded to the redistribution layer structure RDL1 through the bumps **300d**.

[0026] Afterwards, an underfill layer UF1 is formed to fill the space between the redistribution layer structure RDL1 and the semiconductor die **300**, and surrounds the bumps **300d**. In some embodiments, the underfill layer UF1 includes a molding compound such as epoxy, and is formed using dispensing, injecting, and/or spraying process.

[0027] Still referring to FIG. 3, an encapsulation layer E2 is formed around the semiconductor die **300** and the through integrated fan-out vias TIV2. The encapsulation layer E2 may be formed by a molding process followed by a grinding process. In some embodiments, the surface of the semiconductor die **300**, the surfaces of the through integrated fan-out vias TIV2 and the surface of the encapsulation layer E2 are flushed with each other.

[0028] Thereafter, a redistribution layer structure RDL2 is formed over and electrically connected to the semiconductor die **300** and the through integrated fan-out vias TIV2. In some embodiments, the redistribution layer structure RDL2 includes redistribution layers **108** embedded by polymer layers **106**. In some embodiments, each of the redistribution layers **108** includes copper, nickel, titanium, the like, or a combination thereof, and is formed by photolithography, plating, and photoresist stripping processes. In some embodiments, each of the polymer layers **106** includes a polymer material such as polybenzoxazole (PBO), polyimide (PI), benzocyclobutene (BCB), a combination thereof or the like, and is formed by a suitable fabrication technique such as spin-coating, lamination, deposition or the like.

[0029] Afterwards, under bump metallization (UBM) pads **110** are formed over the redistribution layer structure RDL2. In some embodiments, the UBM pads **110** include copper, nickel, titanium, the like, or a combination thereof, and is formed by photolithography, plating, and photoresist stripping processes. The UBM pads **110** are regarded as part of the redistribution layer structure RDL2 in some examples.

[0030] In some embodiments, bumps B are formed over and electrically connected to the UBM pads **110**. The bumps B include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps B may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing. The bumps B are referred to as “ball grid array (BGA) bumps” in some examples.

[0031] In some embodiments, at least one integrated passive device (IPD) **500** is provided and bonded to the redistribution layer structure RDL2 through the bumps **502**. The integrated passive device **500** may include a capacitor, an inductor or a resistor, or the like. For example, the integrated passive device **500** is a capacitor. In some embodiments, the bumps **502** include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps **502** may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing.

[0032] Referring to FIG. 4, the carrier C1 is then removed from the structure formed thereon. In some embodiments, the structure with the carrier C1 is turned over, such that the carrier C1 faces up for the subsequent processes. In some embodiments, the carrier C1 is debonded from the back sides of the semiconductor dies **100** and **200**. In some embodiments, a debonding layer between the carrier C1 and the semiconductor die **200** and between the carrier C1 and the padding layer PL is decomposed under heat of light, and the carrier C1 is then released from the structure formed thereon.

[0033] Referring to FIG. 5, the padding layer PL is removed from the backside of the semiconductor die **100**. The removing process includes an etching process or a laser drilling process. An intermediate structure including two redistribution layer structures and three semiconductor dies is thus formed at this stage. As shown in FIG. 5, upon the removing process of the padding layer PL, a recess R is formed in the encapsulation layer E1 and exposes the back side of the semiconductor die **100**. Specifically, the encapsulation layer E1 has a first portion P1 and a second portion P2 protruded from a top surface of the first portion P1 and a sidewall of the second portion P2 is exposed. In some embodiments, the first portion P1 encapsulates the sidewall of the semiconductor die **100** and the sidewalls of the through integrated fan-out vias TIV1, and the second portion P2 encapsulates the sidewall of the semiconductor die **200**. In some embodiments, the first DAF adhered to the semiconductor die **100** and the second DAF adhered to the semiconductor die **200** may be removed. In other embodiments, the first DAF adhered to the semiconductor die **100** and the second DAF adhered to the semiconductor die **200** may remain as needed.

[0034] Referring to FIG. 6, a semiconductor die **400** is provided and bonded to the through integrated fan-out vias TIV2 through the bumps **402**. The semiconductor die **400** may include a logic die, a memory die, a CPU, a GPU, an xPU, a MEMS die, a SoC die, or the like. In some embodiments, the bumps **402** include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps **402** may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing. The semiconductor dies **100**, **200**, **300** and **400** may have the same or different functions. In some embodiments, each of the semiconductor dies **100**, **200** and **300** is a SoC die, and the semiconductor die **400** is a memory die (e.g., DRAM die). In some embodiments, the surface of the semiconductor die **400** is flushed with the surface of the semiconductor die **200**. In other embodiments, the surface of the semiconductor die **400** may be different from (e.g., lower than or higher than) the surface of the semiconductor die **200**.

[0035] Thereafter, an underfill layer UF2 is formed to fill the space between the semiconductor die **400** and the semiconductor die **200**, and surrounds the bumps **402**. In some embodiments, the

underfill layer UF2 includes a molding compound such as epoxy, and is formed using dispensing, injecting, and/or spraying process. In some embodiments, the underfill layer UF2 partially fills in the gap between the semiconductor die **400** and the second portion P2 of the encapsulation layer E1, as shown in FIG. 6. However, the disclosure is not limited thereto. In some embodiments (not shown), the underfill layer UF2 completely fills the gap between the semiconductor die **400** and the second portion P2 of the encapsulation layer E1. In some embodiments, the underfill layer UF2 and the encapsulation layer E1 include difference materials, so an interface is present between the underfill layer UF2 and the encapsulation layer E1. In other embodiments, the underfill layer UF2 and the encapsulation layer E1 are made by the same material, so an interface is not present between the underfill layer UF2 and the encapsulation layer E1.

[0036] Referring to FIG. 7, a wafer dicing process is performed on the structure of FIG. 6 along the cutting lines CL, so as to cut through the encapsulation layer E1, the redistribution layer structure RDL2, the encapsulation layer E2 and the redistribution layer structure RDL1. After the wafer dicing process or singulation process, the adjacent integrated circuit packages **10** are separated from each other, as shown in FIG. 7. The integrated circuit package **10** of some embodiments is thus completed. In some embodiments, a board substrate such as a printed circuit board (PCB) and/or an interposer substrate such as a silicon interposer or an organic interposer may be provided below and bonded to the integrated circuit package **10** through the bumps B.

[0037] Possible modifications and alterations can be made to the described integrated circuit package **10**. These modifications and alterations are provided for illustration purposes, and are not construed as limiting the present disclosure.

[0038] The integrated circuit package **11** of FIG. 8 is similar to the integrated circuit package **10** of FIG. 7, wherein like reference numerals refer to like elements. The materials, configurations and forming methods of elements of FIG. 7 may refer to those of similar elements described in the previous embodiments.

[0039] The difference between the integrated circuit package **11** of FIG. 8 and the integrated circuit package **10** of FIG. 7 lies in the operation of encapsulating the semiconductor die **400**. In the integrated circuit package **10** of FIG. 7, the semiconductor die **400** is encapsulated by an underfill layer UF2. However, in the integrated circuit package **11** of FIG. 8, the semiconductor die **400** is encapsulated by an encapsulation layer E3.

[0040] Specifically, the encapsulation layer E3 is formed around the semiconductor die **400** and surround the bumps **402**. In some embodiments, the encapsulation layer E3 is in contact with the encapsulation layer E1 and completely fills the gap between the semiconductor die **400** and the second portion P2 of the encapsulation layer E1. The encapsulation layer E3 may be formed by a molding process followed by a grinding process. In some embodiments, the surface of the semiconductor die **400**, the surface of the encapsulation layer E3, the surface of the semiconductor die **200** and the surface of the encapsulation layer E1 are flushed with each other. In some embodiments, the encapsulation layer E3 and the encapsulation layer E1 include difference materials, so an interface is present between the encapsulation layer E3 and the encapsulation layer E1. In other embodiments, the encapsulation layer E3 and the encapsulation layer E1 are made by the same material, so an interface is not present between the encapsulation layer E3 and the encapsulation layer E1.

[0041] The structures of integrated circuit packages of some embodiments are described below with reference to FIG. 7 to FIG. 8.

[0042] In some embodiments, an integrated circuit package **10/11** includes a first redistribution layer structure RDL1, a first semiconductor die **100**, a second semiconductor die **200** and a third semiconductor die **300**. The first redistribution layer structure RDL1 has a first side S1 and a second side S2 opposite to each other. The first semiconductor die **100** and a second semiconductor die **200** are electrically connected to the first side S1 of the first redistribution layer structure RDL1 and encapsulated by a first encapsulation layer E1, wherein the first semiconductor die **100** and the

second semiconductor die **200** have different heights, the first encapsulation layer **E1** has a first portion **P1** and a second portion **P2** protruded from a top surface of the first portion **P1** and a sidewall of the second portion **P2** is exposed, the first portion **P1** encapsulates the first semiconductor die **100**, and the second portion **P2** encapsulates the second semiconductor die **200**. The third semiconductor die **300** is electrically connected to the second side **S2** of the first redistribution layer structure **RDL1** and encapsulated by a second encapsulation layer **E2**.

[0043] In some embodiments, the integrated circuit package **10/11** further includes a fourth semiconductor die **400** disposed on the second portion **P2** of the first encapsulation layer **E1**. In some embodiments, the fourth semiconductor die **400** is separated from the first portion **P1** and the second portion **P2** of the first encapsulation layer **E1**. In some embodiments, a surface of the fourth semiconductor die **400** is flushed with a surface of the second semiconductor die **200**. In some embodiments, a critical dimension of the second semiconductor die **200** is less than a critical dimension of the first semiconductor die **100**.

[0044] In some embodiments, the height **H2** of the semiconductor die **200** is greater than the height **H1** of the semiconductor die **100**. In some embodiments, the height **H2** of the semiconductor die **200** is substantially equal to the total height of the height **H1** of the semiconductor die **100** and the height **H4** of the semiconductor die **400**.

[0045] In some embodiments, the height **H1** of the semiconductor die **100** ranges from about 200 μm to 500 μm , and the height **H2** of the semiconductor die **200** ranges from about 600 μm to 1000 μm , and the height **H4** of the semiconductor die **400** ranges from about 200 μm to 500 μm . In some embodiments, the height **H3** of the semiconductor die **300** ranges from about 200 μm to 500 μm .

[0046] In some embodiments, the integrated circuit package **10/11** further includes a second redistribution layer structure **RDL2** electrically connected to the third semiconductor die **300** and opposite to the first distribution layer structure **RDL1**. In some embodiments, the integrated circuit package **10/11** further includes bumps **B** electrically connected to the second redistribution layer structure **RDL2** and opposite to the third semiconductor die **300**. In some embodiments, the integrated circuit package **10/11** further includes at least one integrated passive device **500** electrically connected to the second redistribution layer structure **RDL2** and opposite to the third semiconductor die **300**.

[0047] The above embodiments in which the semiconductor dies **100** and **200** are formed prior to the formation of the semiconductor dies **300** and **400** are provided for illustration purposes, and are not construed as limiting the present disclosure. In other embodiments, the forming sequence of these semiconductor dies **100** to **400** can be adjusted as needed.

[0048] FIG. **9** to FIG. **13** are cross-sectional views of integrated circuit packages in accordance with some embodiments, wherein like reference numerals refer to like elements. Possible modifications and alterations can be made to the described elements. These modifications and alterations are provided for illustration purposes, and are not construed as limiting the present disclosure. It is understood that the disclosure is not limited by the method described below. Additional operations can be provided before, during, and/or after the method and some of the operations described below can be replaced or eliminated, for additional embodiments of the methods.

[0049] Although FIG. **9** to FIG. **13** are described in relation to a method, it is appreciated that the structures disclosed in FIG. **9** to FIG. **13** are not limited to such a method, but instead may stand alone as structures independent of the method.

[0050] Referring to FIG. **9**, a carrier **C1** is provided. In some embodiments, the carrier **C1** is a non-semiconductor material, such as a glass carrier, a ceramic carrier, or the like. Thereafter, through integrated fan-out vias **TIV2** are formed on the carrier **C1**. In some embodiments, the through integrated fan-out vias **TIV2** include copper, nickel, titanium, the like, or a combination thereof, and are formed by photolithography, plating, and photoresist stripping processes.

[0051] Afterwards, a semiconductor die **300** is placed on the carrier **C1**. In some embodiments, the

semiconductor die **300** is surrounded by the through integrated fan-out vias TIV2. In some embodiments, the semiconductor die **300** is adhered to the carrier C1 with a third die attach film (DAF) between the carrier C1 and the back side of the semiconductor die **300**.

[0052] The semiconductor die **300** may include a logic die, a memory die, a CPU, a GPU, an xPU, a MEMS die, a SoC die, or the like. In some embodiments, the semiconductor die **300** is a SoC die.

[0053] In some embodiments, the semiconductor die **300** includes a substrate **300a**, through substrate vias **300b**, an interconnect structure **300c**, connectors **300d** and a passivation layer **300e**. The substrate **300a** is a silicon substrate. The substrate **300a** has a transistor (not shown) formed thereon, and the interconnect structure **300c** is formed over the substrate **300a** and electrically connected to the transistor. In some embodiments, the substrate **300a** has through substrate vias **300b** (also called “through silicon vias” in some examples) formed therein. In some embodiments, ends of the through substrate vias **300b** are landed on some conductive features of the interconnect structure **300c**, and opposite ends of the through substrate vias **300b** are not revealed from the back surface of the substrate **300a** at this stage. The interconnect structure **300c** includes conductive features embedded by dielectric layers. The conductive features include metal lines, metal vias, metal pads and/or metal connectors. In some embodiments, each conductive feature includes Cu, Al, Ti, Ta, W, Ru, Co, Ni, the like, or a combination thereof. In some embodiments, each dielectric layer includes silicon oxide, silicon nitride, silicon oxynitride, SiOC, the like, or a combination thereof. An etching stop layer may be interposed between two adjacent dielectric layers. The dielectric layers of the interconnect structure **300c** may be replaced by polymer layers or insulating layers as needed. The connectors **300d** are metal pillars (e.g., copper pillars). The metal pillars include Cu, W, Ni, Sn, Ti, Au, an alloy or a combination thereof, and are formed by an electroplating process. The passivation layer **300e** is formed around the connectors **300d**. The passivation layer **300e** includes a polymer material such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), the like, or a combination thereof. In some embodiments, the surfaces of the connectors **300d** are flushed with the surface of the passivation layer **300e**.

[0054] An encapsulation layer E2 is then formed around the semiconductor die **300** and the through integrated fan-out vias TIV2. The encapsulation layer E2 may be formed by a molding process followed by a grinding process. In some embodiments, the surface of the semiconductor die **300**, the surfaces of the through integrated fan-out vias TIV2 and the surface of the encapsulation layer E2 are flushed with each other.

[0055] Referring to FIG. 10, a redistribution layer structure RDL1 is formed over and electrically connected to the semiconductor die **300** and the through integrated fan-out vias TIV2. In some embodiments, the redistribution layer structure RDL1 includes redistribution layers **104** embedded by polymer layers **102**. In some embodiments, each of the redistribution layers **104** includes copper, nickel, titanium, the like, or a combination thereof, and is formed by photolithography, plating, and photoresist stripping processes. In some embodiments, each of the polymer layers **102** includes a polymer material such as polybenzoxazole (PBO), polyimide (PI), benzocyclobutene (BCB), a combination thereof or the like, and is formed by a suitable fabrication technique such as spin-coating, lamination, deposition or the like.

[0056] Thereafter, through integrated fan-out vias TIV1 are formed on the padding layer PL. In some embodiments, the through integrated fan-out vias TIV1 include copper, nickel, titanium, the like, or a combination thereof, and are formed by photolithography, plating, and photoresist stripping processes. In some embodiments, the heights of the through integrated fan-out vias TIV1 are equal to the height of the semiconductor die **100**. However, the disclosure is not limited thereto. In other embodiments, the heights of the through integrated fan-out vias TIV1 are different from (e.g., greater than) the height of the semiconductor die **100**.

[0057] Afterwards, a semiconductor die **100** and a semiconductor die **200** are provided and placed side by side on the redistribution layer structure RDL1. In some embodiments, the semiconductor die **100** is bonded to the redistribution layer structure RDL1 through the bumps B1, and the

semiconductor die **200** is bonded to the redistribution layer structure RDL1 through the bumps B2. [0058] Each of the semiconductor dies **100** and **200** may include a logic die, a memory die, a CPU, a GPU, an xPU, a MEMS die, a SoC die, or the like. The semiconductor dies **100** and **200** may have the same or different functions. In some embodiments, each of the semiconductor dies **100** and **200** is a SoC die.

[0059] In some embodiments, the semiconductor die **100** includes a first silicon substrate and a first device layer formed thereon. The first device layer includes a first transistor, a first interconnect structure, etc., and the bumps B1 are formed over the first silicon substrate and electrically connected to the first device layer. In some embodiments, the bumps B1 include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps B1 may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing.

[0060] In some embodiments, the semiconductor die **200** includes a second silicon substrate and a second device layer formed thereon. The second device layer includes a second transistor, a second interconnect structure, etc., and the bumps B2 are formed over the second silicon substrate and electrically connected to the second device layer. In some embodiments, the bumps B2 include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps B2 may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing.

[0061] The difference between the semiconductor die **100** and the semiconductor die **200** lies in the critical dimensions (e.g., the minimum line width, the minimum hole size, etc.) and chip heights (e.g., the height from the substrate to the connectors or bumps).

[0062] The critical dimension of the semiconductor die **100** is much greater than the critical dimension of the semiconductor die **200**. In some embodiments, the ratio of the critical dimension of the semiconductor die **100** to the critical dimension of the semiconductor die **200** is greater than 1.2, 1.5, 2.0, 2.5 or more. For example, the critical dimension of the semiconductor die **200** ranges from about 5 nm to 10 nm, and the critical dimension of the semiconductor die **200** ranges from about 1 nm to 4 nm. The die with the smallest critical dimension is referred to as a “high-performance die”, a “key die” or a “core die” in some examples. In the disclosure, the high-performance die is the semiconductor die **200** through the specification.

[0063] The height H2 of the semiconductor die **200** is much greater than the height H1 of the semiconductor die **100**. In some embodiments, the ratio of the height H2 of the semiconductor die **200** to the height H1 of the semiconductor die **100** is greater than 1.2, 1.5, 2.0, 2.5 or more. For example, the height H1 of the semiconductor die **100** ranges from about 200 μm to 500 μm , and the height H2 of the semiconductor die **200** ranges from about 600 μm to 1000 μm .

[0064] In some embodiments, the ratio of the second silicon substrate of the semiconductor die **200** to the first silicon substrate of the of the semiconductor die **100** is greater than 1.2, 1.5, 2.0, 2.5 or more. Thicker silicon substrate provides larger thermal mass for the high-performance die (e.g., semiconductor die **200**), and therefore, the heat dissipation of the package is effectively reduced.

[0065] Still referring to FIG. 10, a semiconductor die **400** is provided and placed on the semiconductor die **100**. In some embodiments, the semiconductor die **400** is bonded to the through integrated fan-out vias TIV2 through the bumps 402. The semiconductor die **400** may include a logic die, a memory die, a CPU, a GPU, an xPU, a MEMS die, a SoC die, or the like. In some embodiments, the bumps 402 include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps 402 may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing. The semiconductor dies **100**, **200**, **300** and **400** may have the same or different functions. In some embodiments, each of the semiconductor dies **100**, **200** and **300** is a SoC die, and the semiconductor die **400** is a memory die (e.g., DRAM die). In some embodiments, the surface of the semiconductor die **400** is flushed with the surface of the semiconductor die **200**. In other embodiments, the surface of the semiconductor die **400** may be different from (e.g., lower than or higher than) the surface of the semiconductor die **200**.

[0066] Thereafter, an encapsulation layer E1 is formed around the semiconductor die **100**, the semiconductor die **200**, the semiconductor die **400** and the through integrated fan-out vias TIV1. The encapsulation layer E1 may be formed by a molding process followed by a grinding process. In some embodiments, the surface of the semiconductor die **400**, the surface of the semiconductor die **200**, the surfaces of the through integrated fan-out vias TIV1 and the surface of the encapsulation layer E1 are flushed with each other. In some embodiments, the encapsulation layer E1 surrounds the bumps B1 and B2. However, the disclosure is not limited thereto. In other embodiments, before the operation of forming the encapsulation layer E1, a first underfill layer is optionally formed between the semiconductor die **100** and the redistribution layer structure RDL1 and surrounds the bumps B1, and a second underfill layer is optionally formed between the semiconductor die **200** and the redistribution layer structure RDL1 and surrounds the bumps B2.

[0067] Referring to FIG. **11**, a carrier C2 is bonded to the semiconductor die **200** and the semiconductor die **400**, and the structure with the carrier C2 is turned over such that the carrier C1 faces up for the subsequent processes. In some embodiments, the carrier C2 is a non-semiconductor material, such as a glass carrier, a ceramic carrier, or the like. In some embodiments, the semiconductor die **100** is adhered to the carrier C2 with a first die attach film (DAF) between the carrier C2 and the back side of the semiconductor die **100**, and the semiconductor die **200** is adhered to the carrier C2 with a second die attach film (DAF) between the carrier C2 and the back side of the semiconductor die **200**.

[0068] Thereafter, the carrier C1 is removed from the structure formed thereon. In some embodiments, the carrier C1 is debonded from the back side of the semiconductor die **300**. In some embodiments, a debonding layer between the carrier C1 and the semiconductor die **300** is decomposed under heat of light, and the carrier C1 is then released from the structure formed thereon.

[0069] Afterwards, the substrate **300a** of the semiconductor die **300** is thinned, until the through substrate vias **300b** are exposed from the back side of the semiconductor die **300**. In some embodiments, the thinning process includes a suitable grinding process and/or a polishing process, such as chemical mechanical polishing (CMP) or the like. In some embodiments, the thinning process simultaneously removes portions of the through integrated fan-out via TIV2 and a portion of the encapsulation layer E2. Accordingly, the surface of the semiconductor die **300**, the surfaces of the through integrated fan-out via TIV2 and the surface of the encapsulation layer E2 are flushed with each other.

[0070] Referring to FIG. **12**, a redistribution layer structure RDL2 is formed over and electrically connected to the semiconductor die **300** and the through integrated fan-out vias TIV2. In some embodiments, the redistribution layer structure RDL2 includes redistribution layers **108** embedded by polymer layers **106**. In some embodiments, each of the redistribution layers **108** includes copper, nickel, titanium, the like, or a combination thereof, and is formed by photolithography, plating, and photoresist stripping processes. In some embodiments, each of the polymer layers **106** includes a polymer material such as polybenzoxazole (PBO), polyimide (PI), benzocyclobutene (BCB), a combination thereof or the like, and is formed by a suitable fabrication technique such as spin-coating, lamination, deposition or the like.

[0071] Afterwards, under bump metallization (UBM) pads **110** are formed over the redistribution layer structure RDL2. In some embodiments, the UBM pads **110** include copper, nickel, titanium, the like, or a combination thereof, and is formed by photolithography, plating, and photoresist stripping processes. The UBM pads **110** are regarded as part of the redistribution layer structure RDL2 in some examples.

[0072] In some embodiments, bumps B are formed over and electrically connected to the UBM pads **110**. The bumps B include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps B may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing. The bumps B are referred to as "ball grid array (BGA)

bumps” in some examples.

[0073] In some embodiments, at least one integrated passive device (IPD) **500** is provided and bonded to the redistribution layer structure RDL2 through the bumps **502**. The integrated passive device **500** may include a capacitor, an inductor or a resistor, or the like. For example, the integrated passive device **500** is a capacitor. In some embodiments, the bumps **502** include solder bumps, solder caps formed on metal pillars (e.g., copper pillars), or the like. The bumps **502** may be formed by a suitable process such as evaporation, electroplating, ball drop, or screen printing. [0074] Referring to FIG. **13**, the carrier C2 is then removed from the structure formed thereon. In some embodiments, the structure with the carrier C2 is turned over, such that the carrier C2 faces up for the subsequent processes. In some embodiments, the carrier C2 is debonded from the back sides of the semiconductor dies **200** and **400**. In some embodiments, a debonding layer between the carrier C2 and the semiconductor dies **200** and **400** is decomposed under heat of light, and the carrier C2 is then released from the structure formed thereon. In some embodiments, the first DAF adhered to the semiconductor die **100** and the second DAF adhered to the semiconductor die **200** may be removed. In other embodiments, the first DAF adhered to the semiconductor die **100** and the second DAF adhered to the semiconductor die **200** may remain as needed.

[0075] Thereafter, a wafer dicing process is performed on the structure of FIG. **13** along the cutting lines CL, so as to cut through the encapsulation layer E1, the redistribution layer structure RDL2, the encapsulation layer E2 and the redistribution layer structure RDL1. After the wafer dicing process or singulation process, the adjacent integrated circuit packages **12** are separated from each other, as shown in FIG. **13**. The integrated circuit package **12** of some embodiments is thus completed. In some embodiments, a board substrate such as a printed circuit board (PCB) and/or an interposer substrate such as a silicon interposer or an organic interposer may be provided below and bonded to the integrated circuit package **12** through the bumps B.

[0076] The structure of integrated circuit package of some embodiments are described below with reference to FIG. **13**.

[0077] In some embodiments, an integrated circuit package **12** includes a first redistribution layer structure RDL1, a first semiconductor die **100**, a second semiconductor die **200** and a third semiconductor die **300**. The first redistribution layer structure RDL1 has a first side S1 and a second side S2 opposite to each other. The first semiconductor die **100** and a second semiconductor die **200** are electrically connected to the first side S1 of the first redistribution layer structure RDL1 and encapsulated by a first encapsulation layer E1, wherein the height H1 of the first semiconductor die **100** is less than the height H2 of the second semiconductor die **200**, and the critical dimension of the second semiconductor die **200** is less than the critical dimension of the first semiconductor die **100**.

[0078] In some embodiments, the integrated circuit package **12** further includes a fourth semiconductor die **400** stacked on the first semiconductor die **100**, aside the second semiconductor die **200** and encapsulated by the first encapsulation layer E1.

[0079] In some embodiments, the height H2 of the semiconductor die **200** is greater than the height H1 of the semiconductor die **100**. In some embodiments, the height H2 of the semiconductor die **200** is substantially equal to the total height of the height H1 of the semiconductor die **100** and the height H4 of the semiconductor die **400**.

[0080] In some embodiments, the height H1 of the semiconductor die **100** ranges from about 200 μm to 500 μm , and the height H2 of the semiconductor die **200** ranges from about 600 μm to 1000 μm , and the height H4 of the semiconductor die **400** ranges from about 200 μm to 500 μm . In some embodiments, the height H3 of the semiconductor die **300** ranges from about 200 μm to 500 μm .

[0081] In some embodiments, a surface of the fourth semiconductor die **400** is flushed with a surface of the second semiconductor die **200**. In some embodiments, the integrated circuit package **12** further includes a second redistribution layer structure RDL2 electrically connected to the third semiconductor die **300** and opposite to the first distribution layer structure RDL1. In some

embodiments, the integrated circuit package **12** further includes bumps **B** electrically connected to the second redistribution layer structure **RDL2** and opposite to the third semiconductor die **300**. In some embodiments, the integrated circuit package **12** further includes at least one integrated passive device **500** electrically connected to the second redistribution layer structure **RDL2** and opposite to the third semiconductor die **300**.

[0082] In view of the above, in some embodiments of the present disclosure, with novel arrangement of multiple chips in different levels or tiers, the heat dissipation efficiency is effectively improved by increasing silicon thickness of the high-performance die without increasing the total height of the multi-chip semiconductor package.

[0083] In accordance with some embodiments of the present disclosure, a method of an integrated circuit package includes following operations. A padding layer is formed on a portion of a carrier. A first semiconductor die is placed on the padding layer and a second semiconductor die is placed on the carrier. The first semiconductor die and the second semiconductor die are encapsulated with a first encapsulation layer. A first redistribution layer structure is formed over the first semiconductor die, the second semiconductor die and the first encapsulation layer. A third semiconductor die is placed on the first redistribution layer structure. The third semiconductor die is encapsulated with a second encapsulation layer. A second redistribution layer structure is formed over the third semiconductor die and the second encapsulation layer. The carrier is debonded. The padding layer is removed, and therefore, a recess is formed in the first encapsulation layer.

[0084] In accordance with some embodiments of the present disclosure, an integrated circuit package includes a first redistribution layer structure, a first semiconductor die, a second semiconductor die and a third semiconductor die. The first redistribution layer structure having a first side and a second side opposite to each other. The first semiconductor die and a second semiconductor die are electrically connected to the first side of the first redistribution layer structure and encapsulated by a first encapsulation layer, wherein the first semiconductor die and the second semiconductor die have different heights, the first encapsulation layer has a first portion and a second portion protruded from a top surface of the first portion and a sidewall of the second portion is exposed, the first portion encapsulates the first semiconductor die, and the second portion encapsulates the second semiconductor die. The third semiconductor die is electrically connected to the second side of the first redistribution layer structure and encapsulated by a second encapsulation layer.

[0085] In accordance with some embodiments of the present disclosure, a method of forming an integrated circuit package includes the following operations. A first carrier is provided. A lower semiconductor die is placed on the first carrier. The lower semiconductor die is encapsulated with a lower encapsulation layer. A first redistribution layer structure is formed over the lower semiconductor die. A first semiconductor die and a second semiconductor die are placed side by side on the first redistribution layer and an upper semiconductor die is placed on the first semiconductor die. The first semiconductor die, the second semiconductor die and the upper semiconductor die are encapsulated with an upper encapsulation layer. A second carrier is bonded to the second semiconductor die and the upper semiconductor die. The first carrier is debonded.

[0086] Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

[0087] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate

that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit package, comprising: a first redistribution layer structure having a first side and a second side opposite to each other; a first semiconductor die and a second semiconductor die electrically connected to the first side of the first redistribution layer structure and encapsulated by a first encapsulation layer, wherein the first semiconductor die and the second semiconductor die have different heights, the first encapsulation layer has a first portion and a second portion protruded from a top surface of the first portion and a sidewall of the second portion is exposed, the first portion encapsulates the first semiconductor die, and the second portion encapsulates the second semiconductor die; and a third semiconductor die electrically connected to the second side of the first redistribution layer structure and encapsulated by a second encapsulation layer.
2. The integrated circuit package of claim 1, further comprising a fourth semiconductor die disposed on the second portion of the first encapsulation layer.
3. The integrated circuit package of claim 2, wherein the fourth semiconductor die is separated from the first portion and the second portion of the first encapsulation layer.
4. The integrated circuit package of claim 2, wherein a surface of the fourth semiconductor die is substantially flush with a surface of the second semiconductor die.
5. The integrated circuit package of claim 1, wherein a critical dimension of the second semiconductor die is less than a critical dimension of the first semiconductor die.
6. The integrated circuit package of claim 1, further comprising a second redistribution layer structure electrically connected to the third semiconductor die and opposite to the first distribution layer structure.
7. The integrated circuit package of claim 6, further comprising at least one integrated passive device electrically connected to the second redistribution layer structure and opposite to the third semiconductor die.
8. An integrated circuit package, comprising: a first redistribution layer structure having a first side and a second side opposite to each other; and a first semiconductor die and a second semiconductor die electrically connected to the first side of the first redistribution layer structure, wherein a height of the first semiconductor die is less than a height of the second semiconductor die, and wherein a sidewall and a top of the first semiconductor die are encapsulated by different materials.
9. The integrated circuit package of claim 8, further comprising an encapsulation layer encapsulating the sidewall of the first semiconductor die and a sidewall of the second semiconductor die.
10. The integrated circuit package of claim 8, further comprising: a third semiconductor die electrically connected to the first semiconductor die and encapsulated by an encapsulation layer, wherein the encapsulation layer is in contact with the top of the first semiconductor die.
11. The integrated circuit package of claim 10, wherein a surface of the encapsulation layer is substantially flush with a surface of the second semiconductor die.
12. The integrated circuit package of claim 8, further comprising: a third semiconductor die electrically connected to the first semiconductor die; and an underfill layer disposed between the third semiconductor die and the top of the first semiconductor die.
13. The integrated circuit package of claim 8, further comprising: a third semiconductor die electrically connected to the second side of the first redistribution layer structure and encapsulated

by an encapsulation layer.

14. The integrated circuit package of claim 13, further comprising: at least one integrated passive device electrically connected to the third semiconductor die and opposite to the first semiconductor die.

15. An integrated circuit package, comprising: a first redistribution layer structure having a first side and a second side opposite to each other; a first semiconductor die and a second semiconductor die electrically connected to the first side of the first redistribution layer structure; a first encapsulation layer encapsulating the first semiconductor die and a second semiconductor die; through vias penetrating through the first encapsulation layer and aside the first semiconductor die; and a third semiconductor die disposed over the first semiconductor die and electrically connected to the through vias, wherein a width of the third semiconductor die is greater than a width of the first semiconductor die.

16. The integrated circuit package of claim 15, wherein a surface of the third semiconductor die is substantially flush with a surface of the second semiconductor die.

17. The integrated circuit package of claim 15, wherein the first encapsulation layer is separated from the third semiconductor die.

18. The integrated circuit package of claim 15, wherein the first encapsulation layer further encapsulates a sidewall of the third semiconductor die.

19. The integrated circuit package of claim 15, further comprising a second encapsulation layer encapsulating third semiconductor die and in contact with the first encapsulation layer.

20. The integrated circuit package of claim 15, further comprising: a fourth semiconductor die electrically connected to the second side of the first redistribution layer structure and encapsulated by a second encapsulation layer.
